

Satellite-Terrestrial Coordinated Multi-Satellite Beam Hopping Scheduling Based on Multi-Agent Deep Reinforcement Learning

Zhiyuan Lin¹, Graduate Student Member, IEEE, Zuyao Ni¹, Linling Kuang¹, Member, IEEE, Chunxiao Jiang¹, Fellow, IEEE, and Zhen Huang¹, Member, IEEE

Abstract—Non-geostationary orbit (NGSO) constellations enabled by beam hopping (BH) technology are characterized by wide coverage and high spectrum efficiency. However, how to efficiently schedule multi-satellite beam resources to satisfy the heterogeneous and uneven terrestrial traffic demands remains a huge challenge for satellite operators. This paper proposes a satellite-terrestrial coordinated multi-satellite BH scheduling framework, where the complex multi-satellite BH problem is formulated into a long-term and a short-term subproblems. The long-term subproblem is cell-satellite association problem, which is solved by a low-complexity iterative algorithm executed in network operation control center (NOCC) to minimize the traffic load gap among satellites while considering the interference avoidance. The short-term subproblem is multi-satellite traffic-driven BH problem and we propose a multi-agent deep reinforcement learning (MADRL) architecture where each satellite can cooperatively make real-time BH decisions using the well-trained model by QMIX algorithm to adapt to time-varying and heterogeneous traffic. Simulation results demonstrate that the traffic load gap and network delay have been reduced by 70% and 50% respectively compared with non-load-balancing scheme. Besides, the proposed algorithm outperforms other benchmarks in terms of the network throughput under various traffic load cases and the average network delay is kept within 4 ms. Furthermore, the proposed QMIX-BH can be applied to real-time scheduling since the execution time is less than 1 ms.

Index Terms—Beam hopping, load balancing, resource management, multi-agent deep reinforcement learning, NGSO constellation.

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Zhiyuan Lin is with the Department of Electronic Engineering and Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing 100091, China (e-mail: zy-lin19@mails.tsinghua.edu.cn).

Zuyao Ni, Linling Kuang, Chunxiao Jiang, and Zhen Huang are with Tsinghua Space Center and Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing 100091, China (e-mail: nzy@tsinghua.edu.cn; kll@tsinghua.edu.cn; jchx@tsinghua.edu.cn; huangzhen@tsinghua.edu.cn).

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I. INTRODUCTION

THANKS to global seamless coverage and broadband transmission, non-geostationary orbit (NGSO) constellations have exhibited their promising future, which play the role of supplementing terrestrial network [1]. Recently, due to the advantage of low latency, some mega constellations are growing rapidly such as Starlink, Oneweb and Kuiper, which contains thousands of satellites [2]. As a result, multiple coverage is a distinctive characteristic of NGSO constellations, where terrestrial users can be served by multiple satellites within the minimum elevation range. This feature creates opportunities for cooperative transmission, and significantly improves the flexibility of resource allocation, mobility management and interference management in satellite networks [3]. However, the spatially non-uniform distribution of terrestrial users and the mobility of satellites lead to highly dynamic and uneven traffic demands, which has drawn more attention to advanced resource management techniques for satisfying the heterogeneous and time-varying traffic demands.

Fortunately, multi-beam satellites (MBS) have been applied to extend the flexibility in resource allocation by generating multiple high gain spotbeams. In addition, with the help of fast beamforming ability using phased array antenna, beam hopping (BH) technology has been developed in MBS to deliver satellite services with multi-dimension flexibilities, which has been adopted by Starlink and DVB-S2X standard [4]. Besides, the advantage of beam hopping compared with non-hopped systems has been investigated in [5] and the capacity can be improved greatly by sharing the whole available spectrum between spotbeams. Thus, it is a future trend to introduce BH technology into NGSO constellations, which exploits the time-space-domain flexibility to satisfy the uneven traffic demand. However, how to design effective BH patterns (i.e., deciding the illumination position of spotbeams at each timeslot) in multi-satellite scenario is an unprecedented challenge. On the one hand, interference avoidance between spotbeams is a major issue in BH scheduling, meaning that the interference from intra-satellite and inter-satellite needs to be effectively eliminated by space isolation between spotbeams. On the other hand, the scheduling complexity cannot be ignored, especially in multi-satellite multi-beam systems. The search space of BH patterns increases exponentially with the number of satellites and spotbeams. Moreover, BH scheduling is not just

a simple single objective optimization and several performance metrics should be considered simultaneously such as network throughput, latency, fairness, etc.

As a result, in recent times, BH scheduling problem has been studied by many researchers. An iterative BH scheme based on maximizing signal to interference plus noise (SINR) was proposed in [6], where spotbeams are allocated to the timeslot with maximum SINR. Reference [7] studied a BH scheme using Cuckoo search to maximize the weights of the served users. An optimization and genetic algorithm (GA) was investigated in [5] to solve an efficient beam capacity utilization problem. In order to achieve efficient BH resource allocation in low earth orbit (LEO) satellite networks, while simultaneously satisfying the geostationary earth orbit (GEO) interference avoidance, authors in [8] proposed a joint iterative multi-satellite BH pattern design scheme based on minimizing co-channel interference (CCI) to maximize the minimum traffic satisfaction. To deal with the CCI between adjacent illuminated beams, precoding was introduced in [9], [10], and [11], where cluster-based hopping was designed to satisfy hot-spot demand scenario. Furthermore, the synergy and mutual influence of BH and non-orthogonal multiple access (NOMA) was studied in [12], and a resource allocation problem was formulated with the objective of minimizing the gap between traffic demand and offered capacity. In [13], a machine learning (ML) combined with optimization approach was proposed to reduce the outsized decision space and achieve a rapid BH scheduling. Additionally, aiming at multi-satellite BH scenario, authors in [14] proposed a low-complexity BH algorithm based on load balancing and interference avoidance.

However, all the aforementioned algorithms are based on pre-planning BH scheduling, where the scheduling period contains hundreds of timeslots and the long-period BH patterns depend on historical traffic statistics. The fundamental disadvantage of pre-planning BH scheduling is the assumption of full-buffer traffic (i.e., all users have constant and infinite amount of data for transmission), which restricts the ability to meet burst services and time-varying services [15]. Concretely, when the arrival traffic varies dramatically, pre-planning BH scheduling will lead to waste of resources and high queuing delay. Therefore, for the purpose of more effectively matching the time-varying demand, traffic-driven BH scheduling, which designs BH patterns slot by slot relying on real-time status of traffic queues, has gradually been widely concerned by scholars. In [16], authors proposed a weighted greedy strategy to determine the illumination pattern and an improved GA was introduced to jointly allocate power and bandwidth of spotbeams. Besides, a dynamic BH scheduling strategy based on enhanced artificial bee colony algorithm was designed in [17] to reduce the large solution space and improve the convergence. In addition, [18] proposed a maximal user service weight gain (maxUSWG) resource allocation algorithm where the weight of traffic demand and delay sensitivity is evaluated and resources are allocated to cells with larger weight gain. Similarly, the authors of [19] explored a low-complexity heuristic BH method and an effective Maximum-Weighted Clique (MWC) algorithm for return link BH, which takes into account traffic distribution, inter-beam interference and

channel condition. Since the traffic-driven BH scheduling can be modeled as a Markov Decision Process (MDP), some deep reinforcement learning (DRL) based methods have been studied by several researchers [20], [21], [22], [23]. Reference [20] proposed a DRL-powered GA approach to assist in obtaining the optimal decision when satellite environment changes dynamically. Besides, authors in [21] jointly optimized the BH pattern and beam coverage radius using deep Q-network (DQN) learning to maximize the system throughput. Considering that the action space of satellite grows exponentially with the number of spotbeams, a cooperative multi-agents deep reinforcement learning (MADRL) framework was introduced in [22] for joint optimization of BH patterns and bandwidth allocation.

However, above works neglected two important issues: 1) these studies focus on a single satellite scenario rather than the multi-satellite traffic-driven BH scenario. If each satellite simply performs scheduling independently using the above aforementioned methods, spotbeams from different satellites may interfere with each other due to lack of collaboration, deteriorating the system capacity. 2) If there exists a centralized node, such as network operation control center (NOCC), executes the multi-satellite BH scheduling, the curse of dimensionality in action spaces is intractable. Moreover, communication overheads are required for central node to collect complete information such as the queue and channel status of all terminals, which is not suitable for delay-sensitive applications due to the large spatial scale of satellite network.

Inspired by the above discussions, this paper proposes a two-stage satellite-terrestrial coordinated scheduling framework to solve the complicated multi-satellite BH problem. In the first stage, in order to balance the traffic load among satellites and avoid inter-satellite interference, NOCC decides the connection between satellites and cells according to the perceived traffic demands. In the second stage, a well-known MADRL framework, i.e., QMIX [24], is adopted to achieve the real-time traffic-driven BH scheduling, adapting to the short-term dynamic traffic. Furthermore, we introduce the spatial isolation BH patterns to reduce the size of action space, and the parameter sharing mechanism is used between agents to reduce model overheads and accelerate training. The key contributions and innovations of this paper are summarized as follows:

- Formulation of the complicated multi-satellite BH scheduling problem into two long-term and short-term subproblems, namely the cell-satellite association problem and multi-satellite traffic-driven BH problem. We propose a two-stage satellite-terrestrial coordinated scheduling framework to efficiently solve these two subproblems.
- Presentation of a low-complexity iterative algorithm performed in NOCC to solve the cell-satellite association problem. At first, according to long-term traffic statistics, each cell is greedily assigned to one satellite to ensure inter-satellite load balance as much as possible. Then, an iterative process is proposed to further reduce the load gap between satellites while considering inter-satellite interference avoidance.
- Presentation of a centralized training and distributed execution architecture based on QMIX to

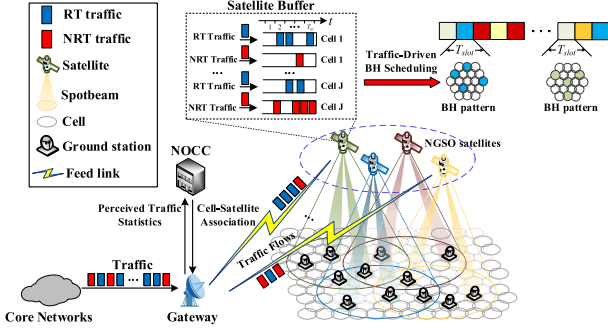


Fig. 1. The architecture of satellite-terrestrial coordinated multi-satellite BH scheduling.

solve the high-dimensional multi-satellite traffic-driven BH problem. Each satellite can distributedly make real-time BH decisions using well-trained model without communication overheads to adapt to the short-term burst traffic. Moreover, spatial isolation BH patterns and parameter sharing mechanism are adopted to reduce the size of action space and model overheads, respectively.

- Presentation of extensive simulation results in terms of long term network throughput and network delay. We demonstrate the superiority of the proposed BH method compared with some benchmark BH schemes. The computational complexity and execution time are provided to demonstrate the validity of real-time BH scheduling.

The rest of the paper is organized as follows. Section II describes the system model and formulates the cell-satellite association problem and multi-satellite traffic-driven BH scheduling problem. Section III provides the solution to the cell-satellite association problem. In Section IV, an advanced QMIX based MADRL algorithm is proposed to solve multi-satellite traffic-driven BH scheduling problem. Simulation results and performance analysis are demonstrated and discussed in Section V, followed by conclusions in Section IV.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

In this paper, we consider the forward link transmission of the multi-beam satellite system consisting of multiple NGSO satellites, as depicted in Fig. 1. There are N satellites serving the ground area which contains a total of M cells. The sets of satellites and cells are denoted by $\mathcal{N} = \{n \mid n = 1, 2, \dots, N\}$ and $\mathcal{M} = \{m \mid m = 1, 2, \dots, M\}$. In particular, the set of cells covered by n -th satellite is represented by $\mathcal{M}_n$, and we assume each satellite can cover J cells (i.e., $|\mathcal{M}_n| = J$, $\forall n \in \mathcal{N}$). Denote the cell index set under the coverage of one satellite as $\mathcal{J} = \{j \mid j = 1, 2, \dots, J\}$. Due to the multiple coverage characteristics of NGSO satellite communication system [8], [25], one cell may be covered by multiple satellites, such that $\mathcal{M} = \bigcup_{n \in \mathcal{N}} \mathcal{M}_n$ and $\mathcal{M}_i \cap \mathcal{M}_j \neq \emptyset, \forall i, j \in \mathcal{N}$.

Due to the non-uniform user distribution, the traffic demands of each cell are much different. Specially, we divide traffic requests into real-time (RT) traffic and non-real-time (NRT) traffic. Denote ϑ_m and ρ_m as the Poisson arrival rate of RT traffic and NRT traffic for m -th cell, respectively.

As shown in Fig. 1, the forward traffic from the core network first arrives at the gateway. Then, the NOCC splits these traffic flows to different satellites according to long-term traffic statistics (i.e., ϑ_m and ρ_m). Each satellite can provide $2 \times J$ buffers to store the arrival RT traffic and NRT traffic for J cells covered by itself. We assume that the buffer for NRT traffic is a First-In, First-Out (FIFO), limited with a maximum length of L_{max} bits, while the buffers for RT traffic can only store the data packets that arrived in the last T_{tll} timeslots, where T_{tll} denotes the Time To Live (TTL) [22], [26]. In other words, the incoming NRT traffic will be dropped if the NRT traffic buffer is full, and the RT data will be discarded if it is not transmitted within T_{tll} timeslots. Let $q_t^{n,j}$ and $d_t^{n,j}$ denote the number of RT and NRT data packets remained in j -th cell's buffers on satellite n by the end of timeslot t respectively. In order to facilitate the calculation of the queuing delay of RT data packet, we further decompose $q_t^{n,j}$ into $\{\phi_{t,1}^{n,j}, \phi_{t,2}^{n,j}, \dots, \phi_{t,l}^{n,j}, \dots, \phi_{t,T_{tll}}^{n,j}\}$, such that $q_t^{n,j} = \sum_{l=1}^{T_{tll}} \phi_{t,l}^{n,j}$ and $\phi_{t,l}^{n,j}$ represents the number of RT data packets which have waited for l timeslots in j -th cell's buffers on satellite n .

In BH system, each satellite can generate K ($K \ll J$) spotbeams simultaneously to serve K different cells in a time division multiple access manner, and each spotbeam can only cover one cell. Denote $\mathcal{K} = \{k \mid k = 1, 2, \dots, K\}$ as the set of spotbeams. Different from the pre-allocated BH scheduling method [9], [10], [11], this paper focuses on traffic-driven BH scheduling, that is, spotbeams of satellites need to be scheduled every timeslot according to the queue status (i.e., $q_t^{n,j}$ and $d_t^{n,j}$). The set of served cells at each timeslot is defined as the BH pattern. Particularly, without loss of generality, the BH pattern of n -th satellite at timeslot t can be expressed as

$$\begin{aligned} X_t^n &= \{x_t^{n,1}, x_t^{n,2}, \dots, x_t^{n,k}, \dots, x_t^{n,K}\}, \\ \text{s.t. } C_1 &: K \leq K_{max}, \\ C_2 &: x_t^{n,k} \in \mathcal{J}, \\ C_3 &: \forall k \neq k' \in \mathcal{K}, x_t^{n,k} \neq x_t^{n,k'}. \end{aligned} \quad (1)$$

C_1 constrains each satellite to generate K_{max} spotbeams at most. C_2 indicates that the k -th spotbeam of satellite n at timeslot t can only irradiate one of the cells that it covers. Besides, C_3 ensures that spotbeams cannot serve the same cell to avoid co-channel interference (CCI).

Assume that each spotbeam could only serve one user located in the cell at each timeslot, and each user has one receiving antenna. A wide-adopted forward channel model is used in this paper [9], [12]. The channel gain from the k -th spotbeam of satellite n to the k' -th user of satellite n' has the form as

$$|h_{nk}^{n'k'}|^2 = \frac{G_{nk,n'k'}^T G_{n,n'k'}^R}{(4\pi \frac{d_{n,n'k'}}{\lambda})^2}, \quad k \in \mathcal{K}, k' \in \mathcal{K}, n \in \mathcal{N}. \quad (2)$$

where $G_{nk,n'k'}^T$ represents the transmitting antenna gain from the k -th spotbeam of satellite n to the k' -th user of satellite n' ; $G_{n,n'k'}^R$ denotes the receiving antenna gain from the user k' of satellite n' to the satellite n ; λ is the wavelength.

Note that $G_{nk,n'k'}^T$ and $G_{n,n'k'}^R$ are determined by the antenna radiation pattern of the satellite and the

user respectively. In particular, the antenna radiation pattern of the satellite refers to ITU-R S.1528 [27], which has the form as

$$G^T(\theta) = \begin{cases} G_{max}^T, & \theta \leq \theta_b, \\ G_{max}^T - 3(\theta/\theta_b)^2, & \theta_b < \theta \leq Y, \\ G_{max}^T + L_s - 25 \log_{10}(\theta/Y), & Y < \theta \leq Z, \\ L_F, & Z < \theta, \end{cases} \quad (3)$$

where θ denotes the off-axis angle in degrees and θ_b is one half the 3 dB beamwidth θ_{3dB} . G_{max}^T represents the maximum gain of transmit antenna, which can be calculated by $G_{max}^T = 10 * \log_{10}[4.93(\frac{70}{\theta_{3dB}})^2]$. $Z = Y \times 10^{0.04(G_{max}^T + L_s - L_F)}$, where $L_s = -6.75$ (dB) denotes the main beam and near-in side-lobe mask cross point below peak gain, $L_F = 0$ (dBi) represents the far-out side-lobe level, and $Y = 1.5\theta_b$.

Besides, the antenna radiation pattern of the user terminal refers to ITU-R S.465-6 [28], which can be expressed as

$$G^R(\theta) = \begin{cases} G_{max}^R, & \theta \leq \theta_{min}, \\ 32 - 25 \log_{10}(\theta), & \theta_{min} < \theta \leq 48^\circ, \\ -10, & 48^\circ < \theta, \end{cases} \quad (4)$$

where $\theta_{min} = \max(2^\circ, 144(D/\lambda)^{-1.09})$ and G_{max}^R denotes the maximum gain of receiving antenna, which has the form as $G_{max}^R = 10 * \log_{10}[(\frac{\pi D}{\lambda})^2]$. D denotes the diameter of user antenna's aperture.

Let P_{nk} denote the transmit power of the spotbeam k belonging to satellite n and we assume that the total power of the satellite P_{sat} is evenly distributed to the spotbeams (i.e., $P_{nk} = P_{sat}/K$) with the purpose of reducing the problem complexity. In order to improve the spectral efficiency of the satellite system, the full frequency reuse is adopted in this paper. Therefore, the signal-to-interference-and-noise ratio (SINR) of the k -th user served by satellite n at timeslot t has the form as

$$\text{SINR}_t^{nk} = \frac{P_{nk}|h_{nk}^{nk}|^2}{\sigma^2 + \sum_{k' \in \mathcal{K} \setminus \{k\}} P_{nk'}|h_{nk'}^{nk}|^2 + \sum_{\substack{n' \in \mathcal{N} \setminus \{n\} \\ k' \in \mathcal{K}_{n'}}} P_{n'k'}|h_{n'k'}^{nk}|^2}, \quad (5)$$

where $\sigma^2 = \kappa T_{\text{noise}} B_w$ represents the noise power. κ , T_{noise} , B_w denotes the Boltzmann's constant, receiver noise temperature and the bandwidth, respectively. It can be observed that the interference in (5) includes two parts. The first term represents the intra-satellite interference while the second term denotes the inter-satellite interference.

Furthermore, according to (1) and (5), the achievable rate of the j -th cell served in satellite n at timeslot t can be expressed as

$$R_t^{n,j} = \sum_{k \in \mathcal{K}} \mathbb{I}(x_t^{n,k} = j) B_w \log_2(1 + \text{SINR}_t^{nk}). \quad (6)$$

$\mathbb{I}(\cdot)$ is the indicator function, indicating that if $x_t^{n,k} = j$, $\mathbb{I}(x_t^{n,k} = j) = 1$; otherwise, $\mathbb{I}(x_t^{n,k} = j) = 0$. It should be noted that there may be multiple spotbeams from different satellites serving the same cell at the same timeslot, resulting in serious inter-satellite interference. Therefore, it is necessary to plan the BH pattern of each satellite reasonably.

B. Optimization Problem Formulation

According to the above system model, we can see that the forward link BH scheduling consists of two stages. The first one is executed at the NOCC to decide which satellites the traffic of each cell should flow to. This flow-split process is a long-term decision due to the slow-varying of perceive traffic statistics. The second one is implemented on each satellite to determine the BH pattern according to the actual arrival traffic, which is a short-term scheduling. Hereafter, we formulate these two sub-problems, namely the cell-satellite association problem and multi-satellite traffic-driven BH problem.

1) *Cell-Satellite Association Problem*: In the cell-satellite association phase, the NOCC needs to decide which cells are connected to which satellites based on the perceived traffic demands at the gateway. This decision requires consideration of two aspects. On the one hand, the total forward traffic flowing to each satellite should be as balanced as possible. Inter-satellite load balancing can make full use of all satellite resources, which avoids the waste of resources caused by some satellites being unloaded and the degradation of Quality of Service (QoS) due to the overloading of some satellites. On the other hand, in order to avoid inter-satellite interference, one cell is not supposed to be associated with multiple satellites unless it is a hotspot cell. Therefore, the cell-satellite association problem is to minimize the traffic load gap among satellites while minimizing the number of multiple served cells, which can be formulated as

$$\begin{aligned} \mathcal{P}_1 : \min_{\{u_{m,n}\}} \Upsilon &= \frac{\Delta}{\Delta_{\text{norm}}} + \mu \frac{\Xi}{\Xi_{\text{norm}}} \quad (\mu \geq 0) \\ \text{s.t. } C_1 : \Delta &= \max\{S_n\} - \min\{S_n\}, \\ C_2 : \Xi &= \sum_{m \in \mathcal{M}} \mathbb{I}(\sum_{n \in \mathcal{N}} u_{m,n} > 1), \\ C_3 : S_n &= \sum_{m \in \mathcal{M}} z_{m,n} u_{m,n}, \\ C_4 : z_{m,n} &= \frac{1}{\sum_{n \in \mathcal{N}} u_{m,n}} (\vartheta_m + \rho_m), \\ C_5 : u_{m,n} &= 0, \quad \forall m \notin \mathcal{M}_n, \\ C_6 : u_{m,n} &\in \{0, 1\}, \quad \forall n \in \mathcal{N}, \quad \forall m \in \mathcal{M}, \end{aligned} \quad (7)$$

where Υ represents the objective function and $u_{m,n}$ is binary variable indicating that whether the cell m is associated with the satellite n . In C_1 , Δ stands for the traffic load gap among satellites, where S_n (denoted by C_3) represents the traffic load assigned to satellite n . Ξ represents the number of multiple served cells, as defined in C_2 . In C_4 , $z_{m,n}$ denotes the traffic demands of m -th cell that the satellite n needs to serve. It can be observed that the total traffic of m -th cell (i.e., $\vartheta_m + \rho_m$) will be evenly allocated to the satellites which are associated with cell m . Δ_{norm} and Ξ_{norm} are normalizing constants.

We introduce a non-negative weight μ into the objective function Υ to adjust the impact of load balancing and inter-satellite interference avoidance. For example, if $\mu = 0$, $\mathcal{P}_1$ only considers load balancing among satellites. Inversely, if μ is set as a large positive number, interference avoidance is prioritized over load balancing, so that each cell must be

assigned to at most one satellite. The solution to $\mathcal{P}_1$ will be further provided in Section III.

2) *Multi-Satellite Traffic-Driven BH Problem*: The objective of this problem is to determine the BH pattern of each satellite every timeslot in a policy that maximizes the total network throughput and minimizes the average delay of RT traffic. Since the BH system serves cells in a TDMA manner and the performance cannot be measured from one timeslot, we should define the long-term metrics, namely long-term network throughput (LTNT) and long-term average delay (LTAD).

Without loss of generality, we focus on the first T timeslots. Denote $\mathcal{T} = \{t \mid t = 1, 2, \dots, T\}$ as the set of timeslots. Considering the high priority characteristics of RT demands, when the j -th cell is being served by satellite n at timeslot t , this satellite will first clear the buffer of RT traffic as much as possible, and then the remaining capacity will transmit NRT data packets if possible. As a result, the throughput of the j -th cell in satellite n at timeslot t can be denoted by

$$\Gamma_t^{n,j} = \min\{R_t^{n,j}T_{\text{slot}}, (q_t^{n,j} + d_t^{n,j})L\}, \quad (8)$$

where T_{slot} denotes the duration of timeslot and L represents the number of bits for each data packet. Note that $R_t^{n,j}T_{\text{slot}}$ is the capacity offered by satellite n for cell j at timeslot t while $(q_t^{n,j} + d_t^{n,j})L$ denotes the total traffic bits that the cell j requires. Thus, the throughput $\Gamma_t^{n,j}$ represents the actual number of bits effectively transmitted.

Now, the LTNT can be expressed as

$$\Gamma = \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}} \sum_{j \in \mathcal{J}} \Gamma_t^{n,j}, \quad (9)$$

which represents the total throughput of the network.

Being ware of the delay sensitivity of RT services, we define the queuing delay of RT data for the j -th cell of satellite n at timeslot t as follow

$$\tau_t^{n,j} = \frac{\sum_{l=1}^{T_{\text{ttl}}} l \cdot \phi_{t,l}^{n,j}}{\sum_{l=1}^{T_{\text{ttl}}} \phi_{t,l}^{n,j}}. \quad (10)$$

Then, the LTAD has the form as

$$\Psi = \frac{1}{|\mathcal{T}||\mathcal{N}||\mathcal{J}|} \sum_{t \in \mathcal{T}} \sum_{n \in \mathcal{N}} \sum_{j \in \mathcal{J}} \tau_t^{n,j}, \quad (11)$$

which represents the average queuing delay of the network.

Since the goal is to simultaneously maximize the total network throughput and minimize the average delay of RT traffic, the total utility function Ω is defined as the weighted sum of LTNT and LTAD defined in (9) and (11), which is denoted by

$$\Omega = (1 - \alpha) \frac{\Gamma}{\Gamma_{\text{norm}}} - \alpha \frac{\Psi}{\Psi_{\text{norm}}}, \quad \alpha \in [0, 1], \quad (12)$$

where Γ_{norm} and Ψ_{norm} are normalization constants to enable the addition of two terms with different units. Besides, α denotes the weighting factor to achieve a balance between LTNT and LTAD.

Therefore, the multi-satellite traffic-driven BH scheduling problem is formulated as a single-objective optimization problem aimed at maximizing the total utility function Ω .

Consequently, the optimization problem is represented as follows,

$\mathcal{P}_2$:

$$\begin{aligned} & \max_{\{X_t^n, \forall n \in \mathcal{N}, \forall t \in \mathcal{T}\}} \Omega \\ & \text{s.t. } C_1 : K \leq K_{\max}, \\ & \quad C_2 : x_t^{n,k} \in \mathcal{J}, \quad \forall n \in \mathcal{N}, \quad \forall k \in \mathcal{K}, \quad \forall t \in \mathcal{T}, \\ & \quad C_3 : x_t^{n,k} \neq x_t^{n,k'}, \quad \forall k \neq k' \in \mathcal{K}, \quad \forall n \in \mathcal{N}, \quad \forall t \in \mathcal{T}. \end{aligned} \quad (13)$$

The optimization variable $X_t^n = \{x_t^{n,1}, x_t^{n,2}, \dots, x_t^{n,K}\}$, represents the BH pattern of satellite n at timeslot t . C_1 constrains the maximum number of spotbeams per satellite. In C_2 , each satellite must serve the cells that it covers. C_3 indicates that spotbeams from one satellite cannot illuminate the same cell in order to avoid intra-satellite interference.

It should be noted that the optimization problem (13) is an interger non-linear programming and non-convex problem, which is well-known to be NP-hard. Besides, due to the dynamic nature of arriving traffic, it is not possible to obtain an optimal solution using integer optimization method such as branch and bound method. Although some other methods such as convex optimization, heuristics and game theory can obtain the sub-optimal solutions, these approaches require global information collection such as the queue status in all satellites and channel state of all cells. Such near-complete information may not be available in real-time for each satellite due to the large propagation delay. Thus, the above approaches are not competent for real-time decision-making in traffic-driven BH scheduling scenario. Fortunately, DRL-based methods can make real-time and near-optimal decisions using trained models without complete information by exploiting special structure such as centralized training and distributed deployment [29]. Therefore, this paper adopts MARL approach to solve the problem formulated in (13), which is detailed Section IV.

III. THE PROPOSED SOLUTION TO CELL-SATELLITE ASSOCIATION PROBLEM

Note that the formulated problem P_1 is integer, non-linear and non-convex due to the presence of the binary variable $u_{m,n}$ and the constraint terms C_2 and C_4 . Hence, it is difficult to obtain the optimal solution in polynomial time, and we propose a low-complexity iterative algorithm to solve P_1 , as depicted in Algorithm 1.

The whole algorithm consists of two steps. The first one is pre-allocation stage. In this process, each cell is only allocated to one satellite, and the goal is to ensure traffic load balance between satellites as much as possible. In particular, we sort the cells in descending order according to total traffic demands, and then assign the cell in turn to its visible satellite with the smallest traffic load. Obviously, this stage is a greedy selection process, and each cell is eventually connected to one satellite.

In the second stage, we start an iterative process, where the cells covered by satellites with larger load are shared to the satellite with smallest load in order to further reduce the load gap between satellites. Specifically, we sort the satellites in

Algorithm 1 An Iterative Solution to P_1

Input: The perceived arrival rate of RT traffic and NRT traffic for each cell ϑ_m and ρ_m , the set of cells covered by the n -th satellite $\mathcal{M}_n$, the number of satellites N , the number of cells M , the weight μ , the maximum number of iterations I_{max} .

Initialize: $\forall n \in \mathcal{N}, S_n = 0, Iter = 0$.

Pre-Allocation Stage:

- 1: Calculate the total perceived traffic demands of each cell:
 $\pi_m = \vartheta_m + \rho_m$.
- 2: Sort cells in descending order of total traffic demands π_m :
 $(m_1, m_2, \dots, m_M) | \pi_{m_1} \geq \pi_{m_2} \geq \dots \geq \pi_{m_M}$.
- 3: **for** $m = m_1$ to m_M **do**
- 4: Obtain the set of visible satellites Θ_m for cell m : $\Theta_m = \{n | m \in \mathcal{M}_n\}$.
- 5: Find the satellite n^* with minimum traffic load from Θ_m : $n^* = \arg \min_{n \in \Theta_m} S_n$.
- 6: Allocate the cell m to the satellite n^* : $u_{m,n^*} = 1$.
- 7: Update $S_n, \forall n \in \mathcal{N}$ according to (7).
- 8: **end for**
- 9: Calculate the objective function value Υ using (7).

Iterative Stage:

- 10: **repeat**
- 11: $\Upsilon_{last} = \Upsilon$.
- 12: Sort satellites in descending order of traffic load S_n :
 $(n_1, n_2, \dots, n_N) | S_{n_1} \geq S_{n_2} \geq \dots \geq S_{n_N}$.
- 13: **for** $n = n_1$ to n_{N-1} **do**
- 14: Find the set of cells $\mathcal{F}$ that are assigned to satellite n but not to satellite n_N .
- 15: **if** $\mathcal{F} = \emptyset$ **then**
- 16: *Continue.*
- 17: **else**
- 18: Calculate the traffic load gap Δ among satellites.
- 19: Find the cell m^* from the set $\mathcal{F}$, which satisfies
 $m^* = \arg \min_{m \in \mathcal{F}} |z_{m,n} - \Delta|$.
- 20: Allocate cell m^* to satellite n_N : $u_{m^*,n_N} = 1$.
- 21: Update $S_n, \forall n \in \mathcal{N}$ according to (7).
- 22: *Break.*
- 23: **end if**
- 24: **end for**
- 25: Calculate the objective function value Υ using (7).
- 26: $Iter = Iter + 1$.
- 27: **until** $\Upsilon > \Upsilon_{last}$ or $Iter \geq I_{max}$.

Output: The cell-satellite allocation results $\{u_{m,n}\}$.

descending order according to their distributed traffic load S_n and calculate the traffic load gap Δ as defined in (7). Next, the set of cells which have been connected to highest-load satellite instead of lowest-load satellite should be determined, denoted by $\mathcal{F}$. Then, we find out the cell from the set $\mathcal{F}$, which satisfies that the traffic it assigns to highest-load satellites closest to Δ . Finally, this cell is allocated to the lowest-load satellite, and we update the objective function value Υ . The above process will be executed in a loop until the maximum number of iterations I_{max} is exceeded or the objective value Υ is no longer reduced.

After finishing the above two steps, the cell-satellite allocation results can be obtained. In other words, which cells each satellite should serve and how much traffic flows to each satellite has been determined. It should be noted that if μ is very large, the iterative process will not be executed since the optimal solution in this case must be that each cell is associated with only one satellite.

IV. THE PROPOSED MARL BASED ON QMIX FOR MULTI-SATELLITE BEAM HOPPING

In this section, we will first present the multi-agent Markov Decision Process (MDP) formulation of multi-satellite traffic-driven BH scheduling. Then, the process of MARL based on QMIX is detailed.

A. Multi-Agent MDP Formulation

In this paper, each satellite is regarded as an agent. Each agent can observe the local state from the environment and take an action (i.e., the BH pattern) at each timeslot. Since all agents work together to maximize the long-term accumulative reward (i.e., the total utility function Ω), it can be considered as a fully cooperative multi-agent task, which can be further represented by a partially observable MDP (POMDP) [30]. In particular, the four key components of POMDP, i.e., state, observation, action, and reward, are defined as follows.

1) *State*: The state of the environment $\mathbf{s}_t$ is the global information about the queue status and channel state of all satellites, which consists of four parts as follows.

- The status of NRT traffic queues with regard to all satellites $\mathbf{D}_t \in \mathbb{R}^{N \times J}$. Specially, $\mathbf{D}_t$ can be further expressed as

$$\mathbf{D}_t = \begin{bmatrix} d_t^{1,1} & d_t^{1,2} & \dots & d_t^{1,J} \\ d_t^{2,1} & d_t^{2,2} & \dots & d_t^{2,J} \\ \vdots & \vdots & \ddots & \vdots \\ d_t^{N,1} & d_t^{N,2} & \dots & d_t^{N,J} \end{bmatrix}, \quad (14)$$

where $d_t^{n,j}$ denotes the queue length of NRT traffic for cell j of satellite n at timeslot t .

- The status of RT traffic queues with regard to all satellites $\mathbf{U}_t \in \mathbb{R}^{N \times J}$. Similarly, $\mathbf{U}_t$ has the form as

$$\mathbf{U}_t = \begin{bmatrix} q_t^{1,1} & q_t^{1,2} & \dots & q_t^{1,J} \\ q_t^{2,1} & q_t^{2,2} & \dots & q_t^{2,J} \\ \vdots & \vdots & \ddots & \vdots \\ q_t^{N,1} & q_t^{N,2} & \dots & q_t^{N,J} \end{bmatrix}, \quad (15)$$

where $q_t^{n,j}$ represents the queue length of RT traffic for cell j of satellite n at timeslot t .

- The queue delay information of RT traffic about all satellites $\Phi_t \in \mathbb{R}^{N \times J}$. In particular, Φ_t can be represented as

$$\Phi_t = \begin{bmatrix} \tau_t^{1,1} & \tau_t^{1,2} & \dots & \tau_t^{1,J} \\ \tau_t^{2,1} & \tau_t^{2,2} & \dots & \tau_t^{2,J} \\ \vdots & \vdots & \ddots & \vdots \\ \tau_t^{N,1} & \tau_t^{N,2} & \dots & \tau_t^{N,J} \end{bmatrix}, \quad (16)$$

where $\tau_t^{n,j}$ is the queuing delay of RT traffic for cell j of satellite n at timeslot t , as defined in (10).

- The channel state of all satellites $\mathbf{H}_t \in \mathbb{R}^{N \times J}$. $\mathbf{H}_t$ can be expressed as

$$\mathbf{H}_t = \begin{bmatrix} h_t^{1,1} & h_t^{1,2} & \dots & h_t^{1,J} \\ h_t^{2,1} & h_t^{2,2} & \dots & h_t^{2,J} \\ \vdots & \vdots & \ddots & \vdots \\ h_t^{N,1} & h_t^{N,2} & \dots & h_t^{N,J} \end{bmatrix}, \quad (17)$$

where $h_t^{n,j}$ is channel gain from satellite n to its j -th cell at timeslot t , which can be obtained by (2).

Finally, the state of the environment is given as $\mathbf{s}_t = [\text{vec}(\mathbf{D}_t); \text{vec}(\mathbf{U}_t); \text{vec}(\mathbf{\Phi}_t); \text{vec}(\mathbf{H}_t)] \in \mathbb{R}^{4NJ \times 1}$, where $\text{vec}(\cdot)$ is the matrix vectorization operation. It should be noted that the state represents the global information, and the agents (i.e., the satellites) cannot obtain this complete information since they only have access to local observations. Therefore, the global state $\mathbf{s}_t$ is only exploited in the centralized trainer.

2) *Observation*: Denote $\mathbf{o}_t^n$ as the local observation accessed by n -th agent at timeslot t . Similar to the state, the local observation $\mathbf{o}_t^n = [\xi_t^n; \chi_t^n; \nu_t^n; \varsigma_t^n] \in \mathbb{R}^{4M \times 1}$, which involves four components as follows:

- The status of NRT traffic queues on the n -th satellite $\xi_t^n = [\xi_{t,1}^n; \xi_{t,2}^n; \dots; \xi_{t,m}^n; \dots; \xi_{t,M}^n] \in \mathbb{R}^{M \times 1}$. The entry $\xi_{t,m}^n$ denotes the cell m 's NRT traffic queue length on satellite n at timeslot t , which has the form as $\xi_{t,m}^n = \sum_{j \in \mathcal{J}} \mathbb{I}(\delta_{n,j} = m) * d_t^{n,j}$. Besides, $\delta_{n,j}$ indicates the corresponding index of the satellite n 's j -th cell in the whole cells set $\mathcal{M}$, and $d_t^{n,j}$ denotes the NRT traffic queue length of j -th cell covered by satellite n . It can be observed that if the cell m is not covered by satellite n , $\xi_{t,m}^n$ will be set as 0. Otherwise, $\xi_{t,m}^n$ is set as the queue length of corresponding cell. Thus, the cell coverage status of satellite n is implied in ξ_t^n .
- The status of RT traffic queues on the n -th satellite $\chi_t^n = [\chi_{t,1}^n; \chi_{t,2}^n; \dots; \chi_{t,m}^n; \dots; \chi_{t,M}^n] \in \mathbb{R}^{M \times 1}$. Similarly, $\chi_{t,m}^n = \sum_{j \in \mathcal{J}} \mathbb{I}(\delta_{n,j} = m) * q_t^{n,j}$ represents the cell m 's RT traffic queue length on satellite n at timeslot t , and $q_t^{n,j}$ denotes the RT traffic queue length of j -th cell covered by satellite n .
- The queue delay of RT traffic regarding the n -th satellite $\nu_t^n = [\nu_{t,1}^n; \nu_{t,2}^n; \dots; \nu_{t,m}^n; \dots; \nu_{t,M}^n] \in \mathbb{R}^{M \times 1}$. Concretely, $\nu_{t,m}^n = \sum_{j \in \mathcal{J}} \mathbb{I}(\delta_{n,j} = m) * \tau_t^{n,j}$ represents the RT traffic queue delay of the cell m on satellite n , and $\tau_t^{n,j}$ is the RT traffic queue delay of j -th cell covered by satellite n .
- The channel information of the n -th satellite $\varsigma_t^n = [\varsigma_{t,1}^n; \varsigma_{t,2}^n; \dots; \varsigma_{t,m}^n; \dots; \varsigma_{t,M}^n] \in \mathbb{R}^{M \times 1}$. In particular, $\varsigma_{t,m}^n = \sum_{j \in \mathcal{J}} \mathbb{I}(\delta_{n,j} = m) * h_t^{n,j}$ represents the channel status from satellite n to cell m , and $h_t^{n,j}$ is the channel gain from satellite n to the j -th cell.

Note that the information in $\mathbf{o}_t^n$ can be accessed by agent n locally. In other words, agents do not have to communicate with each other for information exchange.

3) *Action*: Denote a_t^n as the action of agent n at timeslot t and let $\mathcal{A}_n = \{1, 2, \dots, |\mathcal{A}_n|\}$ represent the actions space of agent n , such that $a_t^n \in \mathcal{A}_n$. At each timestep, the satellite n should decide which K ($K \leq K_{max}$) cells out of J cells to be served. Hence, the number of feasible BH patterns equals $\sum_{k=1}^{K_{max}} \frac{J!}{k!(J-k)!}$, resulting in large action space.

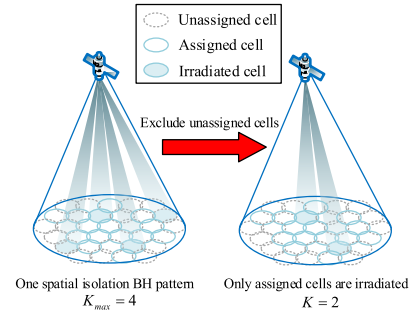


Fig. 2. An illustration of spatial isolation BH pattern and final activated pattern ($K_{max} = 4$, $K = 2$).

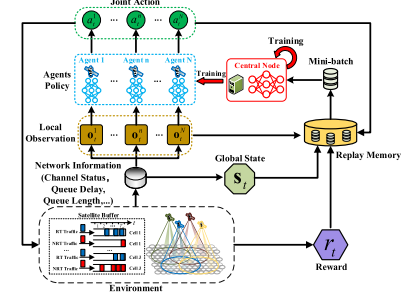


Fig. 3. The overall architecture of MARL based on QMIX.

For the purpose of reducing the size of action space, we introduce the spatial isolation BH patterns as the available BH patterns, where K_{max} spotbeams are not adjacent to each other. Obviously, it is easy to see that $|\mathcal{A}_n| \ll \sum_{k=1}^{K_{max}} \frac{J!}{k!(J-k)!}$. Moreover, as shown in Fig. 2, if there are unassigned cells covered by spotbeams in one spatial isolation BH pattern, only spotbeams corresponding to the assigned cells will be activated eventually. Therefore, the actual number of activated spotbeams can be less than K_{max} .

4) *Reward*: To maximize the optimization target defined in (12), we design the reward at timeslot t as follows

$$r_t = (1 - \alpha) \frac{\sum_{n \in \mathcal{N}} \sum_{j \in \mathcal{J}} \Gamma_t^{n,j}}{\Gamma_{\text{norm}}} - \alpha \frac{\sum_{n \in \mathcal{N}} \sum_{j \in \mathcal{J}} \tau_t^{n,j}}{|\mathcal{N}| |\mathcal{J}| \Psi_{\text{norm}}} - \text{penalty}, \quad (18)$$

where the $\Gamma_t^{n,j}$ and $\tau_t^{n,j}$ represent the throughput and queue delay of the j -th cell in satellite n at timeslot t given by (8) and (10) respectively. Note that we introduce a *penalty* term in the reward to avoid inter-satellite interference, this penalty is proportional to the number of cells illuminated simultaneously. Thus, the reward will be reduced if multiple satellites illuminate the same cell at a timeslot.

B. The Multi-Agent DRL Method Based on QMIX

The problem of multi-satellite traffic-driven BH scheduling in (13) is solved by a popular MARL algorithm named QMIX [24] in this paper, which is based on centralized training and distributed execution. The overview structure of MARL based on QMIX is depicted in Fig. 3. In the centralized training phase, agents firstly obtain the local observation from the environment and then make decisions according to the local

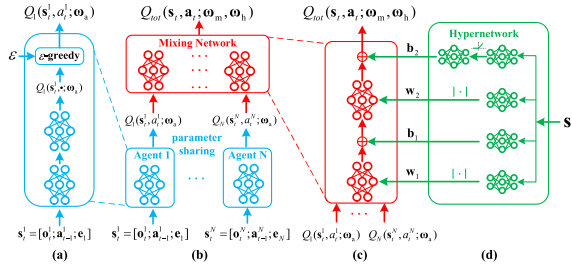


Fig. 4. The architecture of (a) the agent network, (b) the overall QMIX, (c) the mixing network and (d) the hypernetwork.

policy in a distributed manner. Next, the joint action of agents is executed and the reward can be calculated. At each timeslot, the experience such as global state, local observations, joint actions and reward will be stored in the replay memory. Finally, the central node executes centralized training to update the network parameters of agents and itself by exploiting these experiences. In the distributed execution phase, each agent uses the trained model to determine the best BH pattern.

It can be observed that the training is centralized while the execution is decentralized. In other words, the central node can access all local observations and global state, but each agent can make intelligent decisions based on its own local observations without any communications. Next, we will detail the process of offline centralized training and distributed deployment in QMIX.

1) *The Offline Centralized Training*: The overall architecture of QMIX network is depicted in Fig. 4, which consists of agent network, mixing network and hypernetwork. During the training phase, agent n obtains the local observation $\mathbf{o}_t^n$ from the environment, and then put $\mathbf{o}_t^n$, the one-hot encoding of last action $\mathbf{a}_{t-1}^n \in \{0, 1\}^{|\mathcal{A}_n| \times 1}$ and the one-hot encoding of the agent index $\mathbf{e}_n \in \{0, 1\}^{N \times 1}$ (i.e., the n -th element in $\mathbf{e}_n$ equals 1) into the local agent network to output its individual value function $Q_n(s_t^n, a_t^n; \omega_a)$ where $s_t^n = [\mathbf{o}_t^n; \mathbf{a}_{t-1}^n; \mathbf{e}_n]$ represents the final agent network input of agent n and ω_a denotes the parameter of agent network. It should be noted that the parameter sharing mechanism (i.e., all agents sharing one agent network) is adopted in order to promote collaboration among agents. Besides, parameter sharing can speed up the learning and reduce the computation and storage overhead [24].

According to the $Q_n(s_t^n, a_t^n; \omega_a)$, agent n can make a decision using the ε -greedy strategy, which has the form as

$$a_t^n = \begin{cases} \text{random choose } a_t^n \in \mathcal{A}_n, & \text{with probability } \varepsilon, \\ \arg \max_{a_t^n \in \mathcal{A}_n} Q_n(s_t^n, a_t^n; \omega_a), & \text{with probability } 1 - \varepsilon. \end{cases} \quad (19)$$

After each agent chooses an action a_t^n , the reward r_t can be calculated according to (18) and the environment will step into the next state. The central node will collect corresponding value function $Q_n(s_t^n, a_t^n; \omega_a)$ of each agent and the global state s_t . By passing these information into the mixing network and the hypernetwork, the central node will obtain the estimate of joint action-value function $Q_{tot}(s_t, \mathbf{a}_t; \omega_m, \omega_h)$ where $\mathbf{a}_t = (a_t^1, a_t^2, \dots, a_t^N)$ denotes the joint action of agents. ω_m and ω_h represent the parameter of the mixing network and

the hypernetwork respectively. Besides, the experience tuple $\langle s_t, \mathbf{o}_t, \mathbf{a}_t, r_t, s_{t+1}, \mathbf{o}_{t+1} \rangle$ is stored in the replay memory at each timeslot t , where $\mathbf{o}_t = [\mathbf{o}_t^1, \mathbf{o}_t^2, \dots, \mathbf{o}_t^N]$ denotes the joint observation of agents.

In order to guarantee the consistency between the decentralized value functions Q_n and the centralized value function Q_{tot} , QMIX introduces a monotonicity constraint, which has the form as

$$\frac{\partial Q_{tot}(s_t, \mathbf{a}_t; \omega_m, \omega_h)}{\partial Q_n(s_t^n, a_t^n; \omega_a)} \geq 0, \quad \forall n. \quad (20)$$

As shown in Fig. 4, QMIX designs the mixing network and the hypernetwork to enforce the monotonicity (20). The mixing network is a two-layer feed-forward neural network, which takes the outputs of agent network $Q_n(s_t^n, a_t^n; \omega_a)$, $\forall n$ as the input, and mixes them to obtain the joint action-value $Q_{tot}(s_t, \mathbf{a}_t; \omega_m, \omega_h)$. Besides, the parameters of mixing network ω_m are generated by the hypernetwork that is defined as $\mathcal{H}$ with parameter ω_h . Particularly, the global state s_t is fed into the hypernetwork to produce the weights and biases of the mixing network. Let $\mathbf{w}_1, \mathbf{b}_1, \mathbf{w}_2, \mathbf{b}_2$ denote the weights and biases of the first layer and the second layer in mixing network. As a result, we can obtain $\omega_m = (\mathbf{w}_1, \mathbf{b}_1, \mathbf{w}_2, \mathbf{b}_2) = \mathcal{H}(s_t; \omega_h)$.

It should be noted from Fig. 4 that the weights $\mathbf{w}_1$ and $\mathbf{w}_2$ are non-negative since they are generated by a single linear layer, followed by an absolute activation function. Additionally, the first bias $\mathbf{b}_1$ is produced by a single linear layer while the final bias $\mathbf{b}_2$ is produced by two linear layers with a ReLU activation function. As a result, the joint action-value function can be calculated as $Q_{tot}(s_t, \mathbf{a}_t; \omega_m, \omega_h) = \mathbf{w}_2 \cdot f(\mathbf{w}_1 \cdot \mathbf{Q} + \mathbf{b}_1) + \mathbf{b}_2$, where $\mathbf{Q} = [Q_1(s_t^1, a_t^1; \omega_a), \dots, Q_N(s_t^N, a_t^N; \omega_a)]$ and f denotes the ELU activation function. Thus, the derivative of $Q_{tot}(s_t, \mathbf{a}_t; \omega_m, \omega_h)$ with respect to $\mathbf{Q}$ can be obtained by

$$\frac{\partial Q_{tot}(s_t, \mathbf{a}_t; \omega_m, \omega_h)}{\partial \mathbf{Q}} = \mathbf{w}_2 \cdot f'(\mathbf{w}_1 \cdot \mathbf{Q} + \mathbf{b}_1) \cdot \mathbf{w}_1 \geq 0, \quad (21)$$

which is consistent with (20).

This paper adopts the double deep Q-learning (DDQN) method [31] to train all networks. In DDQN, each network has a replica, called the target network. Let ω_a^-, ω_m^- and ω_h^- denote the parameter of target agent network, target mixing network and target hypernetwork respectively.

Finally, the central node trains the networks to minimize the following loss,

$$\mathcal{L}(\omega_a, \omega_m, \omega_h) = \sum_{i \in \mathcal{B}} \left[(y_i^{tot} - Q_{tot}(s_i, \mathbf{a}_i; \omega_m, \omega_h))^2 \right], \quad (22)$$

where $\mathcal{B}$ denotes the set of mini-batch sampled from the replay memory. y_i^{tot} is the target value, which has the form as $y_i^{tot} = r_i + \gamma Q_{tot}(s_{i+1}, \mathbf{a}_{i+1}^*; \omega_m^-, \omega_h^-)$. $\gamma \in [0, 1]$ denotes the discount factor and $\mathbf{a}_{i+1}^* = (a_{i+1}^{1*}, a_{i+1}^{2*}, \dots, a_{i+1}^{N*})$. The entry a_{i+1}^{n*} can be obtained as $a_{i+1}^{n*} = \arg \max_{a_{i+1}^n} Q_n(s_{i+1}, a_{i+1}^n; \omega_a)$.

In this paper, we use Adam [32] optimizer to update the network parameters $\{\omega_a, \omega_m, \omega_h\}$, and the target network

parameter $\{\omega_a^-, \omega_m^-, \omega_h^-\}$ will be copied from $\{\omega_a, \omega_m, \omega_h\}$ every preset timesteps.

2) *The Online Distributed Execution*: Once the offline training is finished, we can deploy this trained model (i.e., the agent network) on each satellite. At each timeslot, each satellite inputs its local observation and determine the optimal BH pattern by choosing the action with the maximum Q-value. It can be seen that this decision-making process does not require the knowledge of other satellites' information. Consequently, the BH pattern can be decided in real time since the computational complexity is much less than traditional algorithms such as convex optimization and metaheuristics. For convenience, a detailed description of centralized training and distributed execution based on QMIX is given in Algorithm 2.

V. SIMULATION RESULTS AND PERFORMANCE ANALYSIS

A. Simulation Parameter

The network parameter setting of QMIX is shown in Table I. The agent network is composed of a gate recurrent unit (GRU) [33] with a fully-connected (FC) layer before and after. All layers in agent network have 64 hidden dimensions. Considering that the action spaces of different agents are inconsistent, the input and output dimensions of the agent network are set to meet the maximum action space among agents when using parameter sharing. The mixing network has a single hidden layer of 64 units. The hypernetwork maps the global state to the weights and biases of mixing network. As shown in Fig. 4, only the final bias of the mixing network is produced by a single hidden layer with 64 units, followed by a ReLU function.

In this paper, a LEO multi-beam satellite system is simulated. Table II summarizes the system parameters. Part of system parameters refer to Starlink and DVB-S2X [4], [34]. There are 9 satellites covering total of 76 cells. Each satellite covers 15 cells and can generate 3 spotbeams at most. The power of satellite is set as 13 dBW and the system bandwidth is 200 MHz. In order to simulate the non-uniform and time-varying traffic, the arrival amount of RT and NRT data for each cell are different and subject to Poisson distribution. Particularly, the Poisson arrival rate of RT traffic for m -th cell, (i.e., ϑ_m), ranges from 20 Mbps to 100 Mbps, while the Poisson arrival rate of NRT traffic for m -th cell, (i.e., ρ_m), ranges from 100 Mbps to 700 Mbps. If cell m is assigned to multiple satellites, ϑ_m and ρ_m will be evenly allocated to each satellite. Moreover, ϑ_m and ρ_m vary every training episode to promote QMIX model to adapt to time-varying traffic demands. The size of each packet is set as 2000 bits and the duration of timeslot is set as 2 ms. Additionally, the TTL of RT traffic is set as 40 timeslots and the maximum queue length of NRT traffic is set as 50 Mbit. The QMIX training parameters are also shown in Table II. The number of training episodes is set as 7000 and each episode contains 128 timesteps. After constant attempts in the early stage, we finally set $\gamma = 0.98$ and the learning rate $\eta = 0.001$ based on the training performance. The replay memory can store the experience of recent 3000 episodes. Since the agent network consists of GRU, we sample batches of 128 episodes experience from the replay buffer and train QMIX every episode. Additionally, the update frequency of target networks is set as 200 episodes.

Algorithm 2 Centralized Training and Distributed Execution Based on QMIX for Multi-Satellite Traffic-Driven Beam Hopping Scheduling

Initialize:

- 1: Initialize networks with random parameter $\{\omega_a, \omega_m, \omega_h\}$, and $\omega_a^- = \omega_a, \omega_m^- = \omega_m, \omega_h^- = \omega_h$.
- 2: Initialize replay memory $\mathcal{D} = \emptyset$.
- 3: Set the learning rate η , the discount factor γ , the exploration probability ε , the target network update frequency F , number of episodes EP , number of steps ST and the mini-batch size $|\mathcal{B}|$.

Offline Centralized Training:

- 4: **for** $episode = 1$ to EP **do**
- 5: Initialize the one-hot encoding of action $\mathbf{a}_0^n = \mathbf{0}, \forall n$.
- 6: **for** $t = 1$ to ST **do**
- 7: Each agent obtains its local observation $\mathbf{o}_t^n$, the one-hot encoding of last action $\mathbf{a}_{t-1}^n$ and the one-hot encoding of the agent index $\mathbf{e}_n$.
- 8: Each agent inputs $\mathbf{s}_t^n = [\mathbf{o}_t^n; \mathbf{a}_{t-1}^n; \mathbf{e}_n]$ into the agent network and chooses the action a_t^n according to ε -greedy strategy as shown in (19).
- 9: Central node collects action-value function $Q_n(\mathbf{s}_t^n, a_t^n; \omega_a)$ of each agent and the current global state $\mathbf{s}_t$ to calculate the joint action-value function $Q_{tot}(\mathbf{s}_t, \mathbf{a}_t; \omega_m, \omega_h)$.
- 10: Execute the joint action $\mathbf{a}_t = (a_t^1, a_t^2, \dots, a_t^N)$ and calculate the reward r_t according to Eq. (18). The environment steps into the next state $\mathbf{s}_{t+1}$ and each agent obtains the next local observation $\mathbf{o}_{t+1}^n$.
- 11: Store the experience tuple $(\mathbf{s}_t, \mathbf{o}_t, \mathbf{a}_t, r_t, \mathbf{s}_{t+1}, \mathbf{o}_{t+1})$ into $\mathcal{D}$.
- 12: Decrease the exploration probability ε .
- 13: **end for**
- 14: **if** $|\mathcal{D}| > |\mathcal{B}|$ **then**
- 15: The central node randomly samples $|\mathcal{B}|$ experiences and calculates the loss $\mathcal{L}(\omega_a, \omega_m, \omega_h)$ according to (22).
- 16: **end if**
- 17: Update the network parameters $\{\omega_a, \omega_m, \omega_h\}$ using the Adam optimizer.
- 18: Update the target network parameters with $\omega_a^- = \omega_a, \omega_m^- = \omega_m, \omega_h^- = \omega_h$ every F episodes.
- 19: **end for**

Online Distributed Execution:

- 20: Each agent loads the trained agent network ω_a .
- 21: Initialize the one-hot encoding of action $\mathbf{a}_0^n = \mathbf{0}, \forall n$.
- 22: **for** $t = 1$ to ST **do**
- 23: Each agent obtains its local observation $\mathbf{o}_t^n$, the one-hot encoding of last action $\mathbf{a}_{t-1}^n$ and the one-hot encoding of the agent index $\mathbf{e}_n$.
- 24: Each agent inputs $\mathbf{s}_t^n = [\mathbf{o}_t^n; \mathbf{a}_{t-1}^n; \mathbf{e}_n]$ into the agent network and chooses the action a_t^n with the maximum $Q_n(\mathbf{s}_t^n, a_t^n; \omega_a)$.
- 25: Agents execute actions and the environment steps into the next state.
- 26: **end for**

TABLE I
PARAMETERS OF QMIX NETWORKS

Layer	Input Size	Output Size	Activation
AGENT NETWORK			
Input Layer (FC)	$4M + \max\{ A_n \} + N$	64	ReLU
RNN (GRU)	64	64	—
Output Layer (FC)	64	$\max\{ A_n \}$	—
MIXING NETWORK			
Input Layer (FC)	N	64	ELU
Output Layer (FC)	64	1	—
HYPERNETWORK			
w_1 Layer (FC)	$4N/J$	$64N$	Abs
b_1 Layer (FC)	$4N/J$	64	—
w_2 Layer (FC)	$4N/J$	64	Abs
b_2 Layer1 (FC)	$4N/J$	64	ReLU
b_2 Layer2 (FC)	64	1	—

TABLE II
SYSTEM PARAMETERS AND QMIX TRAINING PARAMETERS

PARAMETERS	VALUES
SYSTEM PARAMETERS	
Satellite altitude	1500 km
Downlink frequency	11.7 GHz
Number of satellites, N	9
Total number of cells, M	76
Number of cells covered by each satellite, J	15
Maximum number of spotbeams, K_{max}	3
System bandwidth, B_s	200 MHz
Total satellite power, P_{sat}	13 dBW
Satellite 3 dB beamwidth, θ_{3dB}	3°
Maximum transmit antenna gain, G_{max}^T	34.3 dBi
The diameter of user antenna's aperture, D	0.4 m
Maximum receiving antenna gain, G_{max}^R	33.8 dBi
Noise temperature, T_{noise}	300 K
Poisson arrival rate of RT traffic, ϑ_n	20 Mbps ~ 100 Mbps
Poisson arrival rate of NRT traffic, ρ_n	100 Mbps ~ 700 Mbps
Size of data packet, L	2000 bit
Timeslot duration, T_{slot}	2 ms
Time to live of RT traffic in queue, T_{ttl}	40 timeslots
Maximum queue length of NRT traffic, L_{max}	50 Mbit
Normalizing constants, $\Delta_{norm}, \Xi_{norm}, \Gamma_{norm}, \Psi_{norm}$	2000 Mbps, 76, 270 Mbit, 40
Weighting factors, α, μ	$\alpha = 0 : 0.1 : 1, \mu = 0, 1, 10, 100$
QMIX TRAINING PARAMETERS	
Training Episode, EP	7000
Timeslots every episode, ST	128
Learning rate, η	0.001
Replay memory size, $ D $	3000 episodes
Target network update frequency, F	200 episodes
Minibatch size, $ B $	128 episodes
Discount factor, γ	0.98
Initial exploration rate	0.5
Final exploration rate	0.01
Optimizer	Adam

In this paper, we compare the proposed method (called QMIX-BH) against five different BH schemes as follows,

1) *Random Beam Hopping (R-BH)*: In R-BH method, each satellite randomly selects 3 cells out of 15 cells for service at each timeslot.

2) *Periodic Beam Hopping (P-BH)*: P-BH method takes a round-robin beam hopping, which means that 15 cells will be served periodically by the satellite in a fixed polling manner.

3) *Queue Length Prioritized Beam Hopping (QLP-BH)*: In QLP-BH algorithm, each satellite greedily severs 3 cells with the longest queue lengths at each timeslot.

4) *Queue Delay Prioritized Beam Hopping (QDP-BH)*: In QDP-BH algorithm, each satellite greedily severs 3 cells with the highest queue delay at each timeslot.

5) *The Maximal User Service Weight Gain Beam Hopping (USWG-BH)*: USWG-BH was proposed in [18], the cells with the maximum user service weight are served at each timeslot. The weight is determined by the current traffic demand and delay of the cell.

B. Convergence Analysis

In order to illustrate the convergence of the QMIX-BH method, the episode reward and episode loss from one of the training scenarios (i.e., $\alpha = 0.7, \mu = 0$) are depicted in Fig. 5. It can be seen that the reward in Fig. 5(a) increases with the training episodes and gradually converges after 5000 episodes.

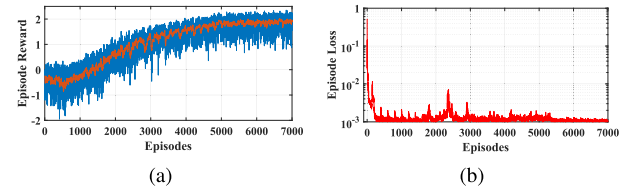


Fig. 5. An illustration of (a) episode reward and (b) loss during training ($\alpha = 0.7, \mu = 0$).

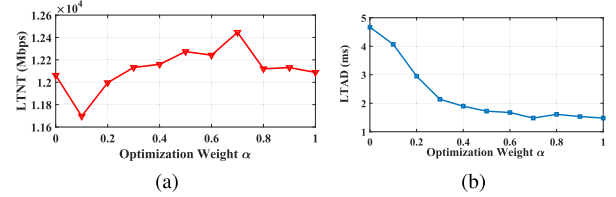


Fig. 6. The performance of (b) LTNT and (c) LTAD under different optimization weight α ($\mu = 0$).

Meanwhile, the episode loss in which Fig. 5(b) remains stable and converges to 10^{-3} after approximately 3000 episodes, which demonstrates that the convergence of QMIX-BH is guaranteed under the parameter setting in Table II. After finishing the offline training, the well-trained QMIX model will be deployed on each agent. Then, we will compare the performance of QMIX-BH with above five BH schemes, and the performance metrics consist of LTNT and LTAD.

C. Performance Analysis

1) Performance Under Different Optimization Weights:

We firstly evaluate the performance of QMIX-BH under different optimization weights α . The Poisson arrival rate of RT traffic ϑ_n is in the range of 20 Mbps to 100 Mbps while the arrival rate of NRT traffic ρ_n is in the range of 100 Mbps to 700 Mbps. Fig. 6 shows the performance of LTNT and LTAD versus α and the results are averaged over 100 experiments. Fig. 6(b) reveals that LTAD decreases with increasing α . This is because that a larger α encourages agents to pay more attention to the cells with high RT traffic, which reduces the queue delay of each cell. Besides, LTAD is less than 2 ms when α is greater than 0.4. However, LTNT does not exhibit monotonicity with α in Fig. 6(a), since LTNT represents the total system throughput depending on the sum of NRT and RT traffic demands. Although priority transmission of RT demands when increasing α , satellites will also transmit NRT services using remaining capacity, resulting in LTNT may not be reduced. In order to achieve a compromise performance between LTNT and LTAD, we choose the trained QMIX model with $\alpha = 0.7$ to compare it with other methods.

2) *Performance Under Different Cell-Satellite Association Schemes*: Next, we evaluate the performance of QMIX-BH under three different cell-satellite association methods as follows:

- **No load balancing scheme**: Each cell is assigned to all satellites it can see. This scheme does not consider the load balance among satellites.
- **Maximum elevation angle scheme**: Each cell is assigned to the satellite which has the maximum elevation angle.
- **Proposed scheme**: This scheme executes Algorithm 1 with $\mu = 0, 1, 10, 100$.

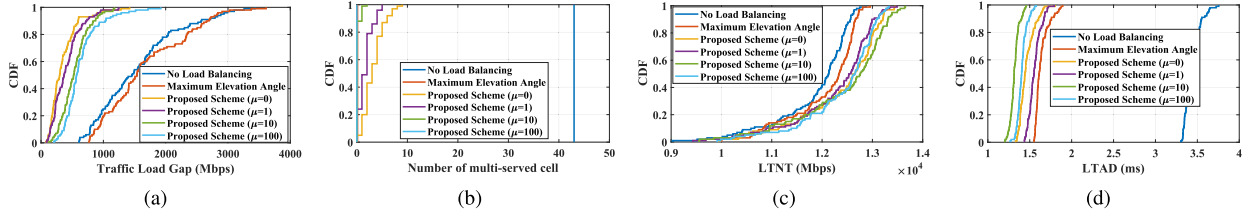


Fig. 7. The performance of (a) traffic load gap, (b) number of multi-served cell, (c) LTNT and (d) LTAD under different cell-satellite association schemes. ($\alpha = 0.7$).

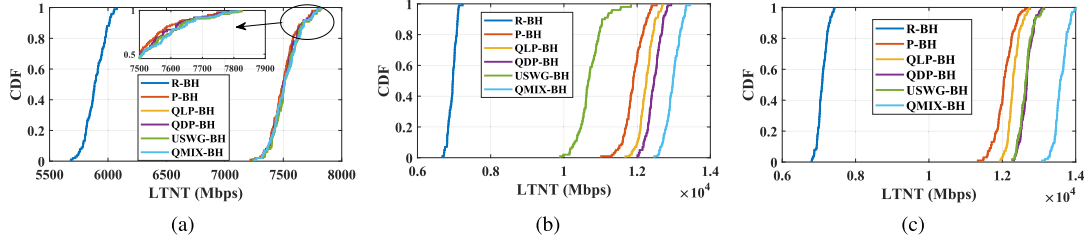


Fig. 8. The cumulative distribution function of LTNT under (a) low demand, (b) medium demand and (c) high demand cases. ($\alpha = 0.7$, $\mu = 10$).

TABLE III
THE AVERAGE TRAFFIC LOAD GAP, NUMBER OF MULTI-SERVED CELL, LTNT AND LTAD OF THREE CELL-SATELLITE ASSOCIATION SCHEMES

Cell-satellite association schemes	Traffic load gap (Mbps)	Number of multi-served cell	LTNT (Mbps)	LTAD (ms)
No load balancing scheme	1553	43	11866	3.44
Maximum elevation angle scheme	1679	0	12057	1.64
Proposed scheme ($\mu = 0$)	357	3.23	12375	1.47
Proposed scheme ($\mu = 1$)	412	1.66	12274	1.56
Proposed scheme ($\mu = 10$)	542	0.13	12427	1.32
Proposed scheme ($\mu = 100$)	641	0	12392	1.41

TABLE IV
THE LIMIT VALUES OF DIFFERENT TRAFFIC LOAD CASES

Load Level	Lower Limit of ϑ_m (Mbps)	Upper Limit of ϑ_m (Mbps)	Lower Limit of ρ_m (Mbps)	Upper Limit of ρ_m (Mbps)
Low Demand	20	50	50	80
Medium Demand	50	100	100	300
High Demand	100	200	400	800

We test on 100 randomly generated traffic instances, and the traffic distribution setting is consistent with the previous subsection. Fig. 7 shows the cumulative distribution function (CDF) of traffic load gap, number of multi-served cell, LTNT and LTAD. Besides, the average results of 100 instances are summarized in Table III. It can be observed that the proposed cell-satellite association method outperforms other schemes because load balancing is considered. In particular, the LTNT has been increased by 300 Mbps $\sim$ 500 Mbps and the LTAD has been reduced by 14% $\sim$ 50%. Besides, as μ increases, the load gap increases while the number of multi-served cell decreases. In particular, when $\mu = 0$, cells may be assigned to multiple satellites to obtain the best load balancing performance, which results in the number of multi-served cell not being equal to 0 and may lead to increased inter-satellite interference. On the contrary, when $\mu = 100$, each cell is only allowed to be connected to one satellite. Thus, there is a loss in load gap performance and the number of multi-served cell equals 0. It is noted that the best LTAD and LTNT performance are obtained with $\mu = 10$ since load balancing and interference avoidance are both considered. Therefore, we use $\mu = 10$ for subsequent simulations.

3) *Performance of Long-Term Network Throughput Under Different BH Schemes:* In this subsection, given the

cell-satellite association scheme (i.e., Algorithm 1 with $\mu = 10$), we compare the LTNT performance of QMIX-BH with five aforementioned BH schemes. In order to evaluate the performance under different traffic loads scenes, ϑ_n and ρ_n are generated from different distributions. In particular, we consider three traffic load cases, i.e., low demand, medium demand and high demand. The limit values of different load types are shown in Table III. We present the simulation results for 100 instances of a given load type. For each instance, ϑ_n and ρ_n are sampled uniformly between the lower limit and upper limit. It should be noted that the limit values are outside the training range, which can verify the generalization of trained QMIX model. Fig. 8 shows the CDF of the LTNT. As depicted in Fig. 8(a), when traffic load is light, all strategies except R-BH show almost similar LTNT around 7500 Mbps since the traffic demands are fully satisfied by the capacity provided by satellites. When traffic requirements increase (e.g., Fig. 8(b) and Fig. 8(c)), QMIX-BH always outperforms other methods, which illustrates that QMIX-BH has better generalization in terms of LTNT. Specially, in high load case, the average LTNT of R-BH, P-BH, QLP-BH, QDP-BH, USWG-BH and QMIX-BH are 7096, 12045, 12307, 12659, 12633 and 13593 Mbps respectively. QMIX-BH has a performance gain of 7% $\sim$ 92% since it learns to improve the network throughput by the spatial isolation BH patterns. Although QLP-BH chooses the cells with most traffic demands to serve, there may be multiple spotbeams illuminating the same cell, deteriorating the throughput performance. In addition, although USWG-BH considers the interference avoidance between spotbeams, it is difficult to design a reasonable weight for the decision attributes of system capacity and delay, which leads to great fluctuations in performance under different traffic loads.

4) *Performance of Long-Term Average Delay Under Different BH Schemes:* In this subsection, given the cell-satellite association scheme (i.e., Algorithm 1 with $\mu = 10$), we evaluate the LTAD performance of different BH algorithms. Fig. 9 shows the CDF of LTAD based on 100 experiments of different traffic load cases. QDP-BH has the lowest LTAD since it serves the cells with highest queue delay

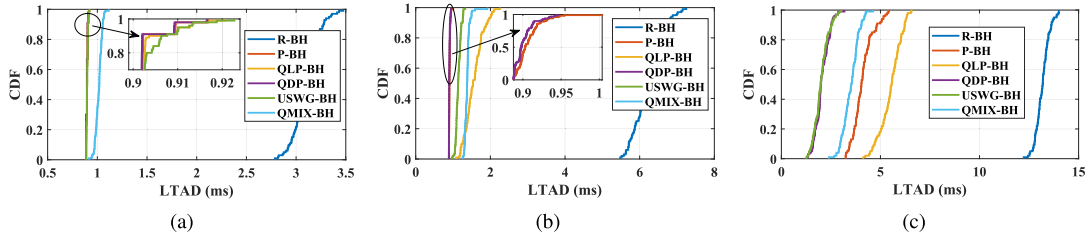


Fig. 9. The cumulative distribution function of LTAD under (a) low demand, (b) medium demand and (c) high demand cases. ($\alpha = 0.7$, $\mu = 10$).

TABLE V
THE AVERAGE LTNT AND LTAD OF DIFFERENT BH SCHEMES

Algorithm	Average LTNT (Mbps)			Average LTAD (ms)		
	Low	Medium	High	Low	Medium	High
R-BH	5892	6955	7096	3.10	6.18	13.18
P-BH	7504	11905	12045	0.90	0.91	4.10
QLP-BH	7525	12254	12307	0.90	1.56	5.52
QDP-BH	7515	12465	12659	0.90	0.90	2.05
USWG-BH	7525	10715	12633	0.90	1.14	2.01
QMIX-BH	7526	12979	13593	1.02	1.38	3.51

at each timeslot. It can be observed from Fig. 9(a) and Fig. 9(b) that LTAD of QMIX-BH is within 1.5 ms in low and medium load scenes. Besides, QMIX-BH slightly losses around 0.1 ms and 0.5 ms performance compared with QDP-BH. However, when traffic demands become heavy such as Fig. 9(c), the performance gap between QMIX-BH and QDP-BH reaches 1.5 ms. This phenomenon can be explained in two aspects. On the one hand, in high load case, QMIX-BH may give priority to serving high-demand cells instead of high-delay cells to better improve the network utility. On the other hand, heavy traffic demands exceed the training range of QMIX, the generalization ability of QMIX-BH may deteriorates in such load case. Finally, the average LTNT and LTAD of different BH algorithms are summarized in Table V for convenience. It should be noted that QMIX-BH maintains the best LTNT performance in any traffic case and the average LTAD has been kept within 4 ms, which is an acceptable delay.

5) *Performance Under Real Network Traces*: In order to verify the proposed algorithm under real network traces, we consider a region with 76 cells in $[112^\circ E, 120^\circ E]$ and $[28^\circ N, 30^\circ N]$, as shown in Fig. 10(a). The total traffic demands are determined according to the population distribution in cells, and the ratio of NRT traffic to RT traffic is 4 : 1. Fig. 10(a) shows the result of cell-satellite association and the pentagram represents the satellite footprint. We can observe that the number of cells allocated to each satellite is different due to the purpose of inter-satellite load balancing. Besides, Fig. 10(b) shows the performance of throughput and delay for each cell. It can be seen that except for a few hotspot cells (e.g., cell 19, 33 and 69), the traffic demands of most cells are well satisfied. In addition, the maximum delay in the cells does not exceed 5 ms, and the average delay remains at 2 ms, which demonstrates that the proposed algorithm can maintain low queue delay while meeting uneven traffic demands.

D. Complexity Analysis

Since the offline trained model is deployed on each satellite, the actual computational complexity of the proposed method consists of the cell-satellite association complexity and model execution complexity, analyzed as follows:

- **The cell-satellite association complexity**: The cell-satellite association problem is solved in Algorithm 1,

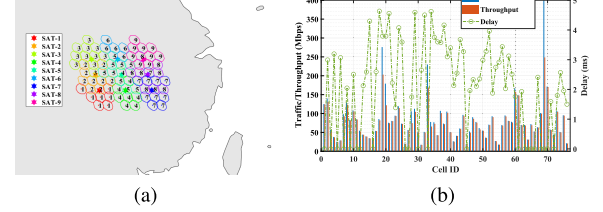


Fig. 10. The result of (a) cell-satellite association and (b) the performance of throughput and delay for each cell in a real scenario. ($\alpha = 0.7$, $\mu = 10$).

TABLE VI
COMPUTATIONAL COMPLEXITY AND EXECUTION TIME
OF PROPOSED BEAM HOPPING ALGORITHM

Stage	Computational Complexity	Execution Time
Cell-Satellite Association	$\mathcal{O}(M \log M + MN + I_{max}(N \log N + NM))$	674 us
Online Model Execution	$\mathcal{O}((4M + 2 \max_n \{ \mathcal{A}_n \} + N + 6N_h)N_h)$	957 us

including pre-allocating stage and iterative stage. The complexity of pre-allocating stage can be represented as $\mathcal{O}(M \log M + MN)$ and the complexity of iterative stage can be expressed as $\mathcal{O}(I_{max}(N \log N + NM))$. Therefore, the overall cell-satellite association complexity is $\mathcal{O}(M \log M + MN + I_{max}(N \log N + NM))$.

- **The model execution complexity**: At each timeslot, each satellite should execute the trained model to make the decision. The complexity of this process depends on the architecture of the agent network. As shown in Table I, agent network includes two FC layers and one GRU. The computational complexity of the first and second FC layer can be represented as $\mathcal{O}((4M + \max_n \{|\mathcal{A}_n|\} + N)N_h)$ and $\mathcal{O}((\max_n \{|\mathcal{A}_n|\})N_h)$ respectively, where N_h denotes the number of hidden units. Besides, the complexity of GRU can be expressed as $\mathcal{O}(N_h(6N_h + 3))$. Thus, the total model execution complexity is around $\mathcal{O}((4M + 2 \max_n \{|\mathcal{A}_n|\} + N + 6N_h)N_h)$.

Finally, the complexity and execution time of the above two parts are summarized in Table VI. It should be noted that the model execution time is around 1 ms, less than the duration of timeslot 2 ms, which indicates that QMIX-BH algorithm can be applied to slot-by-slot beam hopping scheduling.

VI. CONCLUSION

This paper proposes a satellite-terrestrial coordinated multi-satellite BH scheduling framework assisted by QMIX learning, where NOCC determine the cell-satellite association to minimize the traffic gap between satellites and each satellite will make real-time BH pattern decisions using offline-trained

models to maximize the total network utility function. A centralized training and distributed execution architecture is adopted to improve the training efficiency and reduce communication overhead during deployment. For further reduce the size of action space, the spatial isolation BH patterns are introduced and the parameter sharing is used to speed up the learning and reduce the model overhead. Simulation results have proven the superiority of the proposed method in terms of improving the network throughput and reducing the load gap. Meanwhile, the proposed QMIX-BH has short execution time, which can be applied to real-time traffic-driven BH scheduling.

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